

TrES-3b

R-0348

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230611 · created 2026-07-18T09:24:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460106.8505 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.083 +/- 0.052
Transit depth (Rp/Rs)^2	0.69 +/- 0.86 [%]
Orbital Inclination (inc)	83.1 +/- 2.68
Airmass coefficient 1 (a1)	0.872 +/- 0.014
Airmass coefficient 2 (a2)	0.128 +/- 0.076
Scatter in the residuals of the lightcurve fit is	1.62 %
Best Comparison Star	#2 - [190, 403]
Optimal Method	PSF photometry
Transit Duration (day)	0.05 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.083 ± 0.052 vs 0.16309 (deviation 1.51σ)
Duration fit vs published	0.05 d vs 0.0567 d (deviation 0.26σ)
Mid-time O-C	-5.2 min
Depth SNR	1.6
Residual scatter	1.62 %

Light curve

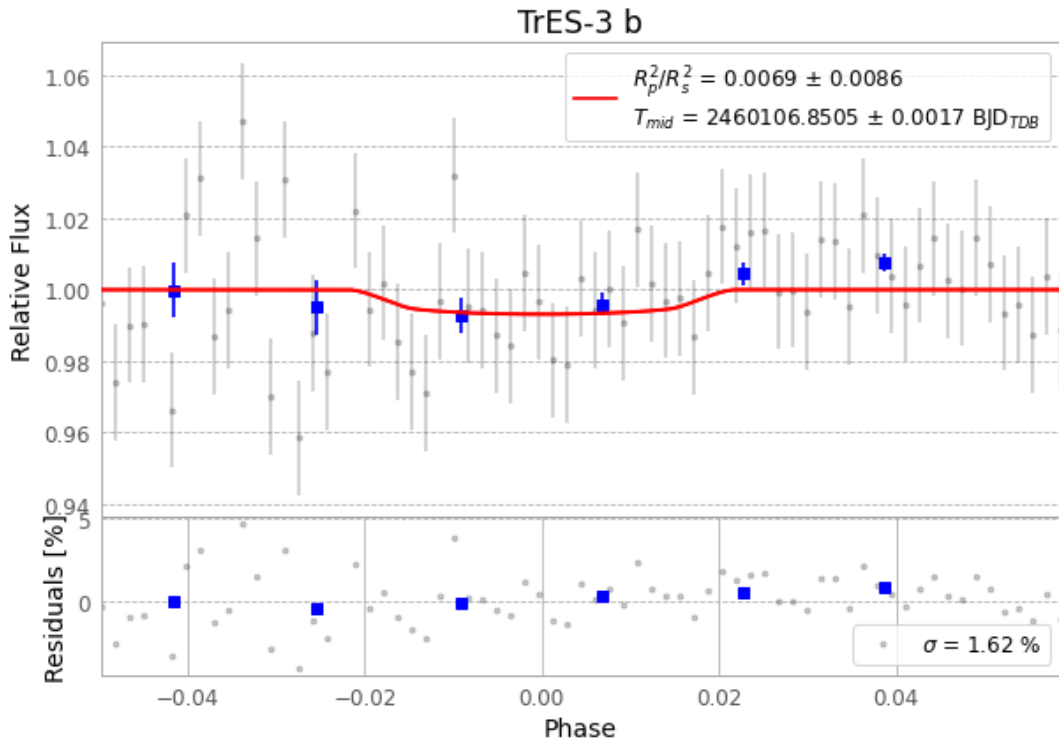


Figure 1 — FinalLightCurve_TrES-3 b_2023-06-10.png (SHA-256 1bb13bcd35d36afa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [260, 317], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 464c30c161687007...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

69 FITS frames (45 MB), TRES-3230611064517.FITS → TRES-3230611100915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230611.log (92e654b241e9...), FinalLightCurve_TrES-3 b_2023-06-10.png (1bb13bcd35d3...), FinalLightCurve_TrES-3 b_2023-06-10.csv (6d99d7fc1ae6...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 09:24Z.